

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 12)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
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6	
7	
8	
9	
10	
11	
TOTAL	

Do not write outside the box

Answer all questions in the spaces provided.

0	1
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Table 1 compares organic and inorganic fertilisers.

0	1
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1

Complete Table 1 by giving two advantages and two disadvantages of each type of fertiliser.

[6 marks]

Table 1

Fertiliser	Advantages	Disadvantages
Organic (manure, sewage sludge, compost)		
Inorganic (eg ammonium nitrate, NPK)		

0	1
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2

Explain how hydroponic cultivation maximises productivity and give **two** disadvantages.

[4 marks]

10

0 2

Table 1 compares an indigenous mixed broadleaf woodland with a Sitka spruce plantation of the same area in Scotland.

Table 1

Feature	Mixed indigenous woodland	Sitka spruce plantation
Tree species	12	1
Age structure	All ages, dead wood present	All planted in the same year
Planting density	Natural, variable	Close, 2 m spacing
Timber yield / $\text{m}^3 \text{ ha}^{-1} \text{ yr}^{-1}$	4	16
Bird species	38	9
Ground flora species	60	4

0 2

1

Use Table 1 to explain why the plantation has a much higher timber yield but far lower biodiversity.

[6 marks]

0 2 . 2 Explain why commercial plantations are felled before the trees reach full size. [2 marks]

0 2 . 3 Explain why there is little demand today for coppiced or pollarded wood. [2 marks]

10

0 3 Table 1 shows characteristics introduced into commercial crops from crop wild relatives (CWRs).

Table 1

Crop	Wild relative source	Characteristic introduced
Sugar cane	Wild sugar cane, Indonesia	Resistance to mosaic virus
Tomato	Single wild species	Resistance to nine major diseases
Rice / barley	Wild salt-marsh varieties	Salt tolerance
Oil palm	Wild palms, central Africa	25% higher yield
Wheat	Wild wheat	Better mycorrhizal associations / phosphate uptake

0 3 . 1 Explain why commercial crop varieties often lack the characteristics shown in Table 1. [3 marks]

03 . 2

Explain what is meant by a **Vavilov centre** (centre of diversity) and why these areas are conservation priorities.

[3 marks]

03 . 3

Explain why a large population does not necessarily have a large gene pool, and why a small gene pool is a problem.

[3 marks]

03 . 4

Give **one** wild plant species with potential as a new commercial food crop and the characteristic that makes it valuable.

[1 mark]

10

04 . 1

Explain how each of the following environmental impacts of salmon farming arises and how it can be reduced.

[8 marks]

Overfishing of wild fish for feed

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Sea lice and disease

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Escapes affecting wild gene pools

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Organic waste pollution

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04 . 2

Explain why culling seals and cormorants near fish farms is controversial and give an alternative.

[2 marks]

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10

0 5 . 1

Which of the following would be most suitable for monitoring the movements of a whale over several months? Tick one box.

[1 mark]

- A Radio collar ☐
- B Acoustic / satellite tag ☐
- C Pitfall trap ☐
- D Geolocator tag recovered by recapture ☐

0 5 . 2

Explain why post-release monitoring is important in a captive breeding and release programme and describe **two** suitable methods.

[4 marks]

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5

0 6 . 1

Describe how each of the following abiotic factors can be managed to increase biodiversity in a nature reserve.

[6 marks]

Light

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Dissolved oxygen

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Nutrients

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06 . 2

Explain how biotic factors can be managed to increase the survival of a species. Give **four** examples.

[4 marks]

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10

07

Table 1 describes three genetically modified (GM) crops.

Table 1

GM crop	Gene source	Characteristic
Bt maize / cotton	Bacillus thuringiensis bacterium	Produces a toxin that kills insect pests
Roundup Ready soya	Agrobacterium	Resistant to the herbicide glyphosate
Golden rice	Daffodil and Erwinia bacterium	Produces vitamin A in the grain

07**1**

Explain the main advantage of GM over conventional selective breeding.

[2 marks]

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07**2**

For each crop in Table 1, explain one potential benefit.

[3 marks]

Bt crops

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Roundup Ready soya

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Golden rice

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07**3**Explain **four** concerns about GM crops.**[4 marks]**

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07 . 4

Give **one** reason why GM crops have been slow to enter cultivation in Europe.

[1 mark]

10

08 . 1

Explain how GPS mapping, drone surveys and computer-based yield monitoring can reduce the environmental impacts of agriculture.

[3 marks]

08 . 2

Explain why productive agricultural technologies are less available in LEDCs.

[2 marks]

08 . 3

Explain how buying fairtrade food affects producers.

[1 mark]

08 . 4

Explain how nitrate from farmland can threaten human health.

[2 marks]

08 . 5 Give **two** agricultural sources of methane.

[2 marks]

1

2

10

09 Table 1 shows some events in the history of life on Earth.

Table 1

Event	Time before present / billion years
Formation of the Earth	4.6
First life (Archaea) on Earth	3.5
First photosynthetic organisms	2.7
Oxygen concentration in the atmosphere starts to rise	2.45
End of the period of oxygen build-up (Proterozoic)	0.54

09 . 1 Use Table 1 to calculate how many **million** years passed between the appearance of photosynthetic organisms and the start of the rise in atmospheric oxygen.

[1 mark]

Answer

million years

09 . 2 Explain why atmospheric oxygen did not start to rise as soon as photosynthesis began.

[2 marks]

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09 . 3

Explain how the rise in atmospheric oxygen made life on the land surface possible.

[3 marks]

09 . 4

Explain how early photosynthetic organisms helped to prevent a long-term rise in the Earth's temperature.

[2 marks]

09 . 5

Give **two** other ways in which living organisms changed the conditions on Earth.

[2 marks]

1

2

10

Describe the conservation efforts used to protect tropical rainforests and evaluate their effectiveness.

ESP/7447/2/SET12

Extra space

10 . 2

State **one** reason why rainforest conservation is important to countries far from the tropics.

[1 mark]

10

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Write an essay on one of the following topics.

1	1	.	1
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Discuss how human activities influence biodiversity, using examples from a range of different contexts.

[25 marks]

OR

1 1 . 2

Discuss how ecological monitoring and the development of new technologies have improved the planning and evaluation of conservation strategies.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

[illegible]

[illegible]

Do not write outside the box

END OF QUESTIONS